Obstacle Avoiding Robot Using Arduino

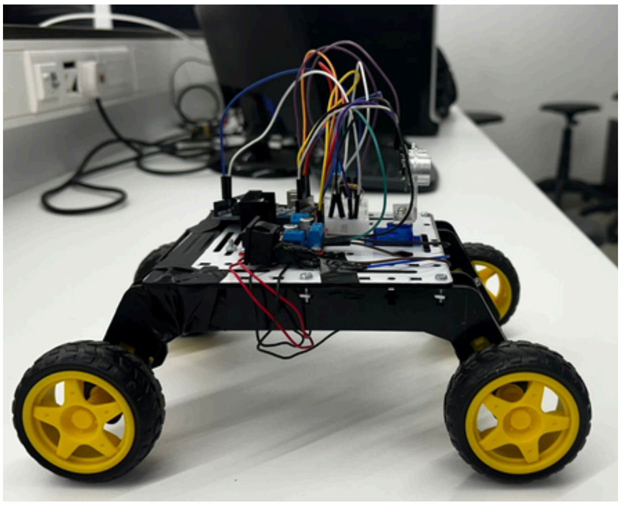
Course: CMBE455 – Embedded Systems

Instructor: Asst. Prof. Dr. Parvaneh Esmaili

Project Description:

An autonomous robot designed to detect and avoid obstacles using an ultrasonic sensor and Arduino microcontroller. The system protects the robot and surrounding objects by reacting in real time without human intervention.

System Design:



Hardware implementation of the obstacle avoiding robot

Code:

```
Start system
Initialize ultrasonic sensor and motors
Loop:
  Read distance twice
  Select minimum distance
  If distance > 25 cm:
     Move forward
  Else:
     Stop robot
     Move backward
     Turn right
```

Project Results & Conclusion:

The robot reliably detects obstacles and avoids collisions in real time, operating autonomously and safely using simple Arduino-based control.

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